VALHEIM WORKERS

THE SETTLEMENT STEWARD'S CODEX

AUTHORITATIVE DOMAIN GUIDE • AUTONOMOUS AI SPECIFICATION • V1.3.0



INSCRIBED FOR THE CHIEFTAINS OF THE TENTH REALM



TEN AUTONOMOUS ROLES

Transform quiet wilderness into bustling strongholds. Every worker autonomously manages tool integrity, material sourcing, crafting pipelines, and defensive patrols.

ZERO-MAINTENANCE ECONOMY

Featuring automated sign bracket routing [Wood], cart freight transport, 50x resource overflow stacking, and intelligent forge repair sequences.

Settlement Operating Architecture

Key Foundations: 

Logistics Arrays: 

Work Post Laws & Boundaries

25-Meter Spherical Operating Post

Each role is bounded to a 25m radius sphere centered on its Work Post totem. Workers evaluate resources, benches, and storage exclusively inside this territory.

Tool Tier vs Material Hardness

Workers automatically scan local containers and their inventory for required tool tiers (e.g., Bronze Pickaxe for Iron, Fine Bow for needle arrows).

10% Durability Repair Cycle

When equipped tool health drops below 10%, the worker immediately halts extraction and navigates to the nearest Forge or Workbench for instant restoration.

Shelter, Rest & Comfort Dynamics

Workers assigned to covered beds gain stamina regeneration buffs and sleep through stormy nights, resuming labor at sunrise.

Codex Chapters & Schemas

Plate I: Master Production Web (p. 3)
Multi-Domain Settlement Macro-Economy • Raw to Refined Flows.

Part I: Lumberjack (p. 4) • Schema 1 (p. 5)
Felling Arcs • Log Bucking • 3m Nursery Grid • 50x Piles.

Part II: Miner (p. 6) • Schema 2 (p. 7)
Subsurface Quarrying • -8m Bedrock • Hoe Planing Ramps.

Part III: Farmer (p. 8) • Schema 3 (p. 9)
1.2m Crop Grid • 50/50 Seed Cycle • Apiary Honey Harvest.

Part IV: Rancher (p. 10) • Herd Tending
Enclosure Feeding • Selective Herd Culling • Byproduct Sourcing.

Part V: Smelter (p. 11) • Schema 4 (p. 12)
High-Heat Pyrolysis • 1:2 Ore/Coal • Ingot Vault Deposition.

Container Routing Syntax [Tags]

[Wood] & [FineWood]

Lumberjacks deposit chopped round logs, core wood, and fine wood into designated chests.

[Ore], [Coal] & [Stone]

Miners dump raw ores and stone chunks; Smelters pull fuel and charge hoppers automatically.

[Produce], [Seed] & [Meat]

Farmers sort vegetables into larders and seeds into planting bins; Ranchers route cuts to cold store.

[Ingot] & [Tools]

Foundrymen store refined bars; Craftsmen pull metals to maintain citadel fortresses.

Expeditions & Citadel Blueprints

Part VI: Cook (p. 13) • Schema 5 (p. 14)
Spit Timers • Cauldron Stews • Fermenter Mead Cellar.

Part VII: Craftsman (p. 15) • Mason & Armorer
Fortress Repair Patrol • Zero-Cost Tool Restorations.

Part VIII: Fisher (p. 16) • Coastal Angler
Shoreline Casting • Tidal Drift • Pier Fillet Station.

Part IX: Hauler (p. 17) • Schema 6 (p. 18)
Ground Vacuum • 1000kg Cart Freight • Sign Bracket Routing.

Part X: Sentry (p. 19) • Schema 7 (p. 20)
Watchtower Defense Geometry • Intersecting Sightlines.

Blueprint Elevations (p. 21) • Diagnostics (p. 22)
Spatial Architecture • 26° Ramps • valheimworkers.cfg.

VALHEIM SETTLEMENT MACRO-LOGISTICS & PRODUCTION WEB

Autonomous Inter-Domain Supply Lines • Circular Resource Flow • Vascular Depot

1. FORESTRY (LUMBERJACK)



Fells trees, bucks timber, excises stumps and re-plants 3m nursery grid.

YIELDS: WOOD

TIMBER TO KILN

5. FOUNDRY (SMELTER & KILN)



Kilns burn wood to coal. Smelters ingest ores (1:2 ratio) into refined bars.

RATIO: 1 ORE : 2 COAL

FUEL & INGOTS

2. QUARRYING (MINER)



Excavates copper, tin & iron veins; planes 26-degree ramps with Hoe.

YIELDS: ORES

RAW ORE

CENTRAL HAULER DEPOT & CART LOGISTICS



Central freight hub. Vacuums 25m drops, hitches 1000kg carts, and sorts to tagged chests [Wood, Ore,

3. AGRICULTURE (FARMER & RANCHER)



Cultivates 1.2m crops; tends apiary hives; feeds & breeds livestock.

50/50 SEED CYCLE

CROPS & MEAT

CAPACITY: 1000 KG

ARROWS & GEAR

4. PROVISIONS (COOK & FISHER)



Roasts meats with 0-char timing; simmers stews; lands tidal catches.

3-SLOT FOOD RATIONS

FOOD RATIONS

6. STRONGHOLD DEFENSE (SENTRY & CRAFTSMAN)



Sentries maintain 40m sightlines; craftsmen repair bastions and tools at 0 cost.

40M SIGHTLINES

SETTLEMENT MACRO-LOGISTICS SPECIFICATIONS & VASCULAR CONDUITS

VASCULAR BASE PRINCIPLES

- 25M POSTS: Overlapping zones ensure zero freight loss.
- CART FREIGHT: 2-wheel wagons replace 6 walking trips.
- 1:2 SMELTER RATIO: 2 Coal burned per 1 Ingot smelted.
- 50X OVERFLOW: Automatic piles prevent chest clogs.
- ZERO-COST REPAIR: [Repair] chests restore gear.
- DIETARY BUFFS: 3-course meals maximize stamina.
- ZERO MOVE PENALTY: 100% material refund on break.
- SENTRY VIGIL: Interlocking 40m sightlines guard camp.
- VACUUM RADIUS: Clears ground debris to save FPS.
- CITADEL WALLS: Craftsmen restore timber defenses.
- DEPOT BALANCING: Equalizes duplicate tagged chests.
- SUSTAINED CYCLE: Never exhausts settlement resources.

AUTONOMOUS AI SCHEDULING & HARVESTS

- NOCTURNAL SLEEP: Civilians sleep 21:00-06:00.
- 1.2M SPACING: Grid prevents crop wither permanently.
- 50/50 SEED CYCLE: Half of crops converted to flowers.
- ZERO-BURN ROASTING: Cook flips meat at 20 seconds.
- 40M SIGHTLINES: Elevated Sentries detect raids early.
- GROUND VACUUM: Clears loose debris within 25m.
- REPAIR PATROL: Craftsmen keep defenses at 100%.
- ROAD BUFF: Carts move 40% faster on stone paths.
- 26° TRANSIT RAMPS: Prevents carts sliding down pits.
- -8M BEDROCK CEILING: Digging halts before world floor.
- 8M PROXIMITY: Chests sit <= 8m from smelter hoppers.
- COLD RESISTANCE: 3-course diet buffers freezing nights.

HUGIN'S LAW: Place Hauler depots directly at the nexus between Extraction (west) and Smelting (east) to eliminate transit bottlenecks.

PART I: THE LUMBERJACK (FORESTER & ARBORIST)

FORESTRY • TIMBER

LOADOUT: 

YIELDS & STATIONS: 

AUTONOMOUS AI WORK-CYCLE

1. ARBORIST PROSPECTING:
Surveys the 25m station boundary, validating equipped axe tier against wood hardness (Birch/Oak need Bronze+).

2. DIRECTIONAL TREE FELLING:
Strikes trunks at ergonomic angles to fell timber into open ground, preventing collateral damage to base roofs.

3. LOG BUCKING & STUMP CLEARING:
Bucks fallen trunks into portable timber chunks and thoroughly excises tree stumps to clear ground.

4. NURSERY REPLANTING GRID:
Deploys Beech seeds, Birch seeds, and Fir cones at 3m spacing to establish perpetual sustainable harvest groves.

ACTIVE DOMAIN CAPABILITIES

[F01] PRECISION TREE FELLING:
Shatters Beech, Birch, Fir, Pine, and Oak trunks using appropriate tool tiers.

[F02] TIMBER LOG BUCKING:
Splits downed logs into regular Wood, Core Wood, and Fine Wood bundles.

[F03] STUMP REMOVAL PASS:
Chops underground stump colliders clean from soil to maximize settlement building area.

[F04] AUTOMATED ARBOR NURSERY:
Re-plants tree seeds in aligned grid patterns to regenerate timber supply within the base.

[F05] 50X WOOD STACK OVERFLOW:
Constructs 50x wood_pile structures automatically when settlement storage boxes are full.

HUGIN'S ADVICE & QUIRKS

STUMP INTEGRITY:
Stumps never regrow. Workers ensure 100% stump clearing so new saplings can be rooted on flat ground.

LOG COLLISION DEFENSE:
Felling massive trees near buildings can damage timber beams; workers path around walls before cutting.

AXE TIER VALIDATION:
Stone axes cannot fell Fine Wood (Birch/Oak). Equip at least a Bronze Axe for advanced biomes.

SETTLEMENT DOMAIN SYNERGY

SMELTER CHARCOAL PIPELINE:
Feeds continuous timber stacks directly to Kilns for unlimited smelting fuel.

CRAFTSMAN TIMBER DEPOT:
Delivers structural Core Wood and Fine Wood directly to the Carpenter for citadel defense.

HAULER WAGON TRANSFER:
Drops heavy lumber stacks into parked wagons for mass player transit.

SCHEMA 1: FORESTRY & ARBOR NURSERY WORK-CYCLE

Directional Felling • 100% Stump Removal • 3m Sapling Grid • 50x Stacks

STEP 01: AXE TIER VALIDATION

Scans nearby trees; checks hardness. Stone Axe for Beech/Fir; Bronze+ for Birch/Oak.

HARDNESS: 1 VS 2

STEP 02: DIRECTIONAL FELLING

Strikes trunk at oblique angles to fell trees into meadows, saving roof beams.

RAYCAST CLEARANCE

STEP 03: LOG BUCKING PASS

Chops fallen trunks into round wood, fine wood, and core wood chunks.

TRUNCATION PASS

STEP 04: 100% STUMP CLEARING

Stumps never regrow! Worker excises root taproot colliders to clear soil for saplings.

TAPROOT EXCISION

STEP 05: 3.0M NURSERY GRID

Deploys Beech, Birch & Fir seeds with exact 3m equidistant spacing to prevent decay.

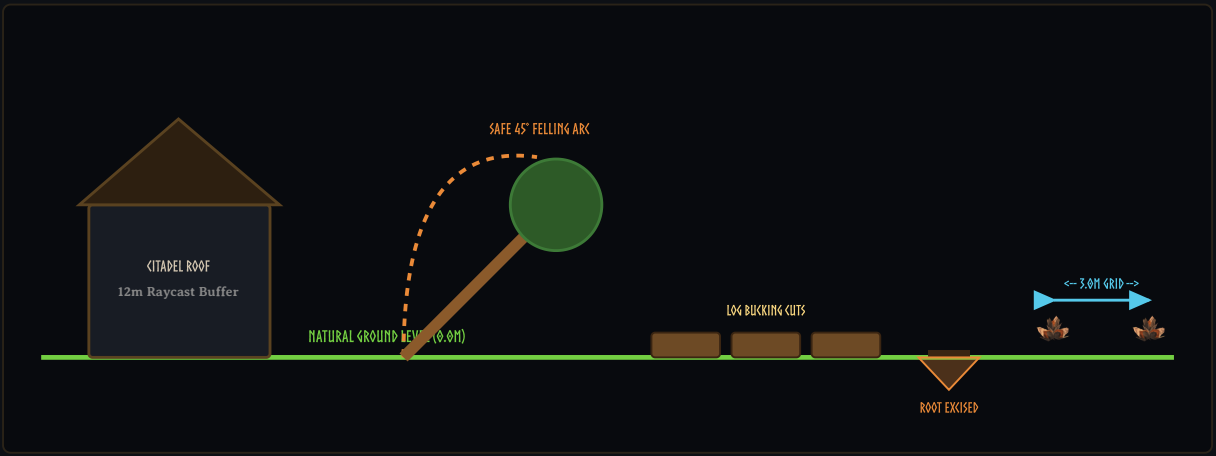
3.0M CLEARANCE

STEP 06: STORAGE & 50X PILES

Sorts logs into [Wood] chests. If storage is full, builds physical 50x stacks on soil.

50X OVERFLOW PREFAB

ARBORIST GEOMETRY & SUSTAINABLE NURSERY BLUEPRINT



FORESTRY LOGISTICS MATRIX

- F01 PRECISION FELLING: Trajectory raycasts check 12m for roofs.
- F02 LOG BUCKING: Splits trunks into regular, core, and fine wood.
- F03 STUMP ELIMINATION: Completely clears root taproot colliders.
- F04 SUSTAINABLE NURSERY: 3m equidistant sapling matrix.
- F05 WOOD PILE OVERFLOW: Builds physical 50x stacks when capped.
- TOOL TIER VALIDATION: Bronze Axe required for Birch & Oak trees.
- ZERO LOG ROT: Freshly harvested timber never decays on soil.
- 25M WORK POST: Bounds all felling activity to designated sphere.
- SEED CYCLE: Automatically recovers beech seeds & cones.

CIRCUIT CONDUITS & SYNERGIES

- SMELTER KILN FEED: Supplies raw wood directly for charcoal pyrolysis.
- CRAFTSMAN TIMBER DEPOT: Delivers Core Wood for fortress log poles.
- FINEWOOD ARSENAL: Delivers fine timber for Huntsman and Finewood bows.
- HAULER CART LOGISTICS: High-volume lumber transferred by wagon.
- FORGE REPAIR RUN: Retires axe to camp Forge when durability < 10%.
- NOCTURNAL REST: Resumes cutting at 06:00 sunrise after shelter sleep.
- STUMP INTEGRITY: Never leave stumps; they block new growth.
- CONTAINER TAG: Directs round logs into chests tagged [Wood].
- FINEWOOD TAG: Directs birch/oak timber into chests tagged [Finewood].

STUMP CLEARING LAW: Never leave stumps in the ground. Stumps prevent nursery saplings from taking root and permanently block terrain pathing.

PART II: THE MINER (QUARRYMAN & EXCAVATOR)

MINING • EXTRACTION

LOADOUT: 

YIELDS & STATIONS: 

AUTONOMOUS AI WORK-CYCLE

1. GEODE PROSPECTING:
Scans 25m sphere around Work Post; evaluates pickaxe tier vs rock hardness.

2. OVERBURDEN EXCAVATION:
Chips away surface crags and topsoil to expose buried ore colliders.

3. HOE PLANING PASS:
Swaps to Hoe to flatten jagged pit bottoms, creating ramped cart paths.

4. FORGE REPAIR RUN:
Navigates to camp Forge when pickaxe durability drops below 10%.

ACTIVE DOMAIN CAPABILITIES

[F06] SURFACE QUARRYING:
Shatters surface boulders, loose rocks, and flint nodes with pickaxe.

[F07] DEEP VEIN EXCAVATION:
Extracts Copper, Tin, Iron Scrap, and Silver veins with tool tier checks.

[F08] EVEN FIELD (HOE LEVELING):
Equips Hoe to eliminate steep pit traps and build smooth transit roads.

[F05] 50X OVERFLOW MASONRY PILES:
Constructs 50x stone_pile structures when camp storage chests are full.

HUGIN'S ADVICE & QUIRKS

PROXIMITY INTEGRITY:
Place an upgraded Workbench and Forge near the quarry to avoid walking delays.

SIGN BRACKET ROUTING:
Mark foundry strongboxes with [Stone] and [Ore] for automated sorting.

BEDROCK BOUNDARY:
Miners safely stop downward digging at the world bedrock floor (-8m).

SETTLEMENT DOMAIN SYNERGY

FOUNDRY FEEDER:
Direct conduit of raw copper, tin, and scrap iron to the Smelter quadrant.

MASON STONE SUPPLY:
Feeds infinite stone blocks to Craftsmen building stronghold ramparts.

HAULER ORE RUNNERS:
Transfers heavy ore loads from subterranean pits up to the surface.

SCHEMA 2: SUBSURFACE QUARRYING & HOE PLANING

Overburden Excavation • -8m Bedrock Limits • 26° Hoe Ramps • 50x Stone Piles

STEP 01: 25M GEODE SCAN

Scans 25m sphere around Work Post; evaluates pickaxe tier vs rock hardness.

SPHERE: 25M

STEP 02: OVERBURDEN CLEARING

Chips away overlying soil & crags to expose contiguous ore colliders.

BEDROCK BOUNDARY

STEP 03: DEEP VEIN EXCAVATION

Extracts raw Copper, Tin, Iron & Silver. Collects ore into cargo packs.

TIER VALIDATION

STEP 04: HOE PLANING PASS

Swaps to Hoe; levels jagged pit floor, eliminates traps, and shapes 26° ramps.

26° TRANSIT RAMP

STEP 05: 10% FORGE REPAIR RUN

Tool integrity safety trigger: when pickaxe durability < 10%, routes to Forge.

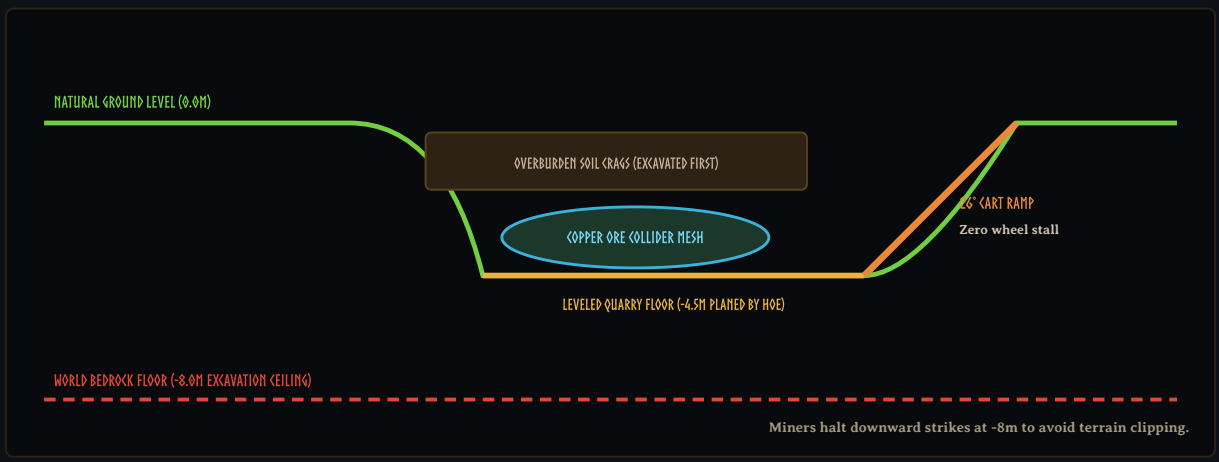
RETREAT: DUR <= 10%

STEP 06: STORAGE & 50X PILES

Sorts ores to [Ore] & [Stone] chests; constructs 50x stone piles on soil if full.

50X PREFAB MASONRY

SUBSURFACE QUARRY CROSS-SECTION & -8M BEDROCK CEILING



Miners halt downward strikes at -8m to avoid terrain clipping.

QUARRY SPECIFICATIONS & LIMITS

- F06 SURFACE QUARRYING: Shatters field boulders & crags.
- F07 DEEP VEIN MINING: Extracts Copper, Tin, Iron, and Silver.
- F08 HOE PLANING: Levels pit bottoms into smooth roadways.
- F05 50X STONE PILES: Automatically places stone stacks on soil.
- BEDROCK HARD CEILING: Downward mining constrained to -8m.
- PICKAXE HARDNESS: Bronze Pickaxe needed for Iron Scrap.
- 10% FORGE REPAIR: Automatically returns to Forge when damaged.
- 25M WORK POST: Spherical boundary prevents unbounded wandering.
- ZERO PIT TRAPS: Planing pass removes steep vertical edges.

HAULER & FOUNDRY CONDUITS

- 26° TRANSIT GRADIENT: Steeper ramps cause carts to stall.
- HAULER EXTRACTION LINK: Empties quarry chests directly.
- FOUNDRY HOPPER PIPELINE: Transfers raw ore to Smelters.
- MASON STONE SUPPLY: Supplies stone blocks to Craftsmen.
- PROXIMITY RULE: Place Forge near pit to minimize walking.
- SIGN BRACKETS: Tag chests [Ore] and [Stone] for auto-sort.
- ZERO COLLAPSE HAZARD: Undermining terrain never collapses.
- FLINT SOURCING: Gathers shoreline flint nodes within boundary.
- SILVER PROSPECTING: Mines mountain silver veins with Iron pick.

CART RAMP LAW: Always construct 26° ingress ramps into deep quarry pits using a Hoe. Steep cliff walls trap haulers and delay ore transit.

PART III: THE FARMER (CULTIVATOR & APIARIST)

LOADOUT: 

YIELDS & STATIONS: 

AUTONOMOUS AI WORK-CYCLE

1. SOIL CULTIVATION:
Tills wild grass into rich fertile loam using the Cultivator tool.

2. 1.2M GRID PLANTING:
Plants seeds in disciplined geometric matrices to eliminate cramped crop wither.

3. FULL-CYCLE HARVESTING:
Harvests fully mature crops and seed-flowers at the perfect botanical peak.

4. APIARY HONEY TENDING:
Empties camp beehives into honeycomb stores; monitors bee happiness.

ACTIVE DOMAIN CAPABILITIES

[F09] TILL SOIL CULTIVATION:
Converts raw terrain into fertile farmland across the full 25m boundary.

[F10] 1.2M SPACING SEEDING:
Plants carrots, turnips, onions, barley, and flax at calculated intervals.

[F11] MATURE VEGETABLE HARVEST:
Gathers ripe food without damaging adjacent growing saplings.

[F12] SEED-FLOWER ROTATION:
Re-seeds mature crops into flowers to guarantee infinite seed generation.

[F13] APIARY HONEY EXTRACTION:
Collects sweet honey combs from all beehives within the post perimeter.

HUGIN'S ADVICE & QUIRKS

GRID SPACING (1.2M):
Crops planted too close wither and die. Workers enforce minimum 1.2m spacing.

SEED RE-INVESTMENT:
Farmers reserve 50% of harvests for re-planting to keep farms permanently self-sustaining.

FENCE ENCLOSURE:
Surround agricultural plots with stakewalls to protect crops from Greydwarf trampling.

SETTLEMENT DOMAIN SYNERGY

CAULDRON PANTRY CONDUIT:
Feeds continuous root vegetables and honey to the Cook for stamina stews.

RANCH TAME FEEDER:
Delivers root vegetables to boars and barley to lox enclosures automatically.

LOOM & SPINNING WHEEL:
Supplies flax directly to Craftsmen for linen thread and padded armor.

SCHEMA 3: AGRICULTURAL 1.2M GRID

SCHEMA 3: AGRICULTURAL 1.2M GRID & APIARY

1.2m Botanical Spacing • 50% Seed Reinvestment • Beehive Proximity • Larder Storage

STEP 01: SOIL CULTIVATION

Converts raw grassy terrain into rich fertile loam using Cultivator.

SPHERE: 25M

STEP 02: 1.2M GRID PLANTING

Deploys seeds in disciplined 1.2m grid to eliminate cramped crop wither.

1.2M SPACING

STEP 03: FULL-CYCLE HARVEST

Harvests fully mature crops and seed flowers at peak botanical ripeness.

ZERO WITHER

STEP 04: SEED ROTATION PASS

Re-seeds 50% of harvest into seed-flowers to guarantee infinite seeds.

50/50 RATIO

STEP 05: APIARY HONEY PASS

Empties camp beehives into honeycomb stores; monitors bee happiness.

4 HONEY / HIVE

STEP 06: LARDER STORAGE

Routes produce into [Produce] chests for kitchen culinary feasts.

TAG: [PRODUCE]

BOTANICAL 1.2M MATRIX & APIARY TERRITORY BLUEPRINT



AGRICULTURAL SPECIFICATIONS

- F09 SOIL CULTIVATION: Tills grass into fertile loam.
- F10 1.2M GRID PLANTING: Eliminates cramped crop wither.
- F11 HARVEST PASS: Gathers mature food without plant harm.
- F12 SEED FLOWER ROTATION: 50% harvest re-seeded into flowers.
- F13 APIARY HARVEST: Gathers honey from all beehives within 25m.
- BIOME CONSTRAINTS: Barley & Flax require Plains soil.
- ZERO FOOD WITHER: Crops never decay once fully grown.
- CULTIVATOR REPAIR: Routes to Forge when durability < 10%.
- RAIN BENEFIT: Rain accelerates botanical hydration.

CIRCUIT CONDUITS & PANTRY LINKS

- CAULDRON CONDUIT: Feeds turnips, onions & honey to Cook.
- RANCH TAME FEEDER: Supplies boars with carrots & lox with barley.
- TAILOR WEAVE SUPPLY: Routes raw flax to spinning wheels.
- BEE HAPPINESS LAW: Beehives require 4m clear open air radius.
- STAKEWALL PERIMETER: Enclose crops to stop boar trampling.
- HAULER CONDUIT: Haulers vacuum vegetable drops into cellars.
- STORAGE TAGS: Mark chests [Produce], [Seed], and [Honey].
- SEED CONSERVATION: Never consumes seeds for cooking recipes.
- NOCTURNAL REST: Resumes seeding at 06:00 sunrise.

BOTANICAL DENSITY LAW: Always space crops at least 1.2m apart. Crops planted too close immediately wither, yielding zero produce.

PART IV: THE RANCHER (HERDSMAN & ANIMAL TAMER)

LIVESTOCK • BREEDING

LOADOUT: 

YIELDS & STATIONS: 

AUTONOMOUS AI WORK-CYCLE

1. ENCLOSURE PATROL:
Inspects pens for Boars, Wolves, and Lox; checks hunger and breeding status.

2. SELECTIVE TROUGH FEEDING:
Deposits carrots, berries, and meats to maintain active reproduction buffs.

3. SUSTAINABLE CULLING:
Spares breeding pairs while culling excess mature beasts for hides and meats.

4. BYPRODUCT HARVEST:
Gathers dropped leather scraps, pelts, wolf fangs, and trophy horns.

ACTIVE DOMAIN CAPABILITIES

[F14] DOMESTIC HERD FEEDING:
Dispenses vegetables to boars, meat to wolves, and barley/cloudberries to lox.

[F15] BEAST TAMING ACCELERATION:
Soothes frightened wild animals to accelerate the domestic taming cycle.

[F16] SELECTIVE HERD CULLING:
Harvests surplus livestock when pen capacity exceeds sustainable limits.

[F17] LEATHER & PELT SOURCING:
Collects hides, entrails, and fangs, routing them into tailoring chests.

HUGIN'S ADVICE & QUIRKS

BREEDING MINIMUM:
Workers always preserve at least 2 breeding adults per enclosure to prevent herd extinction.

ENCLOSURE ROOFING:
Roofed pens protect domestic animals from Bat raids and drake frost breath.

FEED CONDUIT:
Place a chest tagged [Feed] with carrots and meat near pens for autonomous trough loading.

SETTLEMENT DOMAIN SYNERGY

BUTCHER TO COOK PIPELINE:
Provides raw boar, wolf, and lox meats to the kitchen hearth.

CRAFTSMAN TANNERY:
Supplies constant pelts and leather scraps for armors, bows, and banners.

FARMER BYPRODUCT EXCHANGE:
Recycles excess crop yields into animal feed, closing the agricultural loop.

PART V: THE SMELTER (FOUNDRYMAN & PYROTECHNIC)

METALLURGY • INGOTS

LOADOUT: 

YIELDS & STATIONS: 

AUTONOMOUS AI WORK-CYCLE

1. CHARCOAL KILN PYROLYSIS:
Charges kilns with raw timber; produces black coal at high thermal efficiency.

2. SMELTER HOPPER CHARGING:
Feeds raw ores and coal into hoppers in strict 1:2 stoichiometric ratios.

3. BLAST FURNACE REFINING:
Processes Black Metal scrap and Flametal ore in high-tier blast furnaces.

4. INGOT CONSOLIDATION:
Extracts molten bars and deposits them into sorted foundry vaults.

ACTIVE DOMAIN CAPABILITIES

[F18] CHARCOAL KILN OPERATION:
Continuously loads wood to produce charcoal fuel autonomously.

[F19] SMELTER FUEL CHARGING:
Monitors combustion chambers, keeping coal levels consistently topped off.

[F20] ORE REDUCTION & SMELTING:
Feeds copper, tin, scrap iron, and silver ores into active smelters.

[F21] BLAST FURNACE METALLURGY:
Operates advanced blast furnaces for black metal and endgame alloys.

[F22] INGOT VAULT STACKING:
Collects cooled metal bars and routes them to secure vaults.

HUGIN'S ADVICE & QUIRKS

STORAGE PROXIMITY:
Keep raw ore chests within 8m of smelters to minimize worker hopper loading travel.

COAL BUFFERING:
Run 2 Kilns per Smelter to guarantee furnaces never stall for lack of fuel.

CONTAINER TAGGING:
Tag output chests [Ingot] or [Metal] to enable automatic foundry consolidation.

SETTLEMENT DOMAIN SYNERGY

MINER ORE INGESTION:
Consumes raw copper, tin, iron, and silver hauled directly from local quarries.

CRAFTSMAN ARMORY FORGE:
Feeds refined bronze, iron, and black metal bars directly into weapon forges.

LUMBERJACK FUEL TIE-IN:
Consumes surplus wood stacks for kiln combustion, freeing depot capacity.

SCHEMA 4: METALLURGICAL PYROLYSIS

SCHEMA 4: HIGH-HEAT PYROLYSIS & METALLURGY

Kiln Combustion • 8m Supply Radius • 1:2 Ore/Coal Stoichiometry • Ingot Vaults

STEP 01: KILN PYROLYSIS



Loads 25 wood into Charcoal Kiln; burns timber into 25 coal chunks.

25 WOOD = 25 COAL

STEP 02: 8M CHEST PROXIMITY



Pulls raw ores and coal from supply boxes within 8m. Eliminates delays.

PROXIMITY: <= 8.0M

STEP 03: 1:2 STOICHIOMETRY



Maintains combustion ratio: 1 Ore chunk : 2 Coal chunks loaded into hoppers.

COMBUSTION BALANCE

STEP 04: BLAST FURNACE REFINE



Refines high-tier Black Metal scrap & Flametal with high-temperature blast fuel.

BLACK METAL REFINING

STEP 05: SLAG TAPPING RUN



Collects cooling molten metal bars directly from run-off troughs into carts.

BAR EXTRACTION

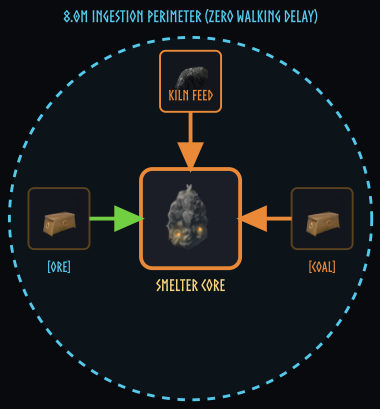
STEP 06: [INGOT] VAULT DEPOSIT



Consolidates finished bronze, iron & silver bars into fortified vaults.

TAG: [INGOT]

FOUNDRY QUADRANT BLUEPRINT & 8-METER INGESTION CONDUIT



METALLURGICAL BENCHMARKS

- BRONZE REFINERY: Requires 2x Copper + 1x Tin at Forge.
- SCRAP IRON: Swamp Crypt mud scraps refined in Smelter.
- SILVER INGOTS: Mountain veins smelted into frost-gear.
- BLACK METAL: Requires high-temp Blast Furnace stoking.
- FOUNDRY PAIRING: 2 Kilns per 1 Smelter stops fuel delay.
- SMELTER CAPACITY: Max 20 Ore | Max 25 Coal | ~30s tick.
- ZERO METAL DESPAWN: Refined bars persist forever on ground.
- AUTOMATED TAPPING: Bars eject directly into cargo carts.
- FUEL BALANCE: Smelter never accepts wood directly; only coal.

LOGISTICS & STORAGE ARCHITECTURE

- 8M RADIUS LAW: Chests outside 8m cause walking delays.
- DIRECT FORGE CONDUIT: Bars route straight to blacksmiths.
- HAULER CART DROP: Raw ore wagons unload directly in yard.
- TIMBER SINK: Consumes surplus wood piles for unlimited coal.
- SMOKE VENTILATION: High chimney required to stop suffocation.
- NOCTURNAL SHUTDOWN: Kilns burn down safely during night.
- TAG MATCHING: Mark output chests [Ingot] or [Metal].
- HOPPER FILL LOGIC: Re-fills immediately when space opens.
- BLAST HOPPER: Artisan Table unlocks blast furnace automation.

FOUNDRY PROXIMITY LAW: Keep raw ore and coal chests within 8m of smelter hoppers. Greater distances waste worker stamina in transit.

PART VI: THE COOK (CHEF & FERMENTER)

PROVISIONS • MEAD

LOADOUT: 

YIELDS & STATIONS: 

AUTONOMOUS AI WORK-CYCLE

1. HEARTH SPIT MONITORING:
Places meats on cooking stations; flips at 20s to prevent scorched charcoal.

2. CAULDRON STEW CRAFTING:
Combines vegetables, honey, and meats into high-tier stamina and HP feasts.

3. STONE OVEN BAKING:
Bakes bread dough, lox meat pies, and fish wraps in stone bread ovens.

4. MEAD CELLAR AGING:
Brews mead bases and ages them in oak fermenters for 2400 seconds.

ACTIVE DOMAIN CAPABILITIES

[F23] COOKING SPIT ROASTING:
Roasts boars, deer, and wolf meats with zero burn-loss failure rate.

[F24] CAULDRON RECIPE SIMMERING:
Prepares stews, soups, sausages, and serpent stews autonomously.

[F25] STONE OVEN BAKERY:
Bakes pies, bread loaves, and gourmet fish dishes in wood-fired ovens.

[F26] FERMENTER MEAD BARRELS:
Brews healing, stamina, frost resistance, and poison resistance meads.

[F27] LARDER CONSOLIDATION:
Sorts rations into cold storage chests tagged with [Food] and [Mead].

HUGIN'S ADVICE & QUIRKS

SPIT BURN PREVENTION:
Worker AI monitors food timers down to the tick, pulling cooked meats immediately.

FERMENTER TIMERS:
Mead takes 2 Valheim days (2400s) to ferment. Keep 3 fermenters rotating continuously.

FOOD ROTATION:
Stock larder chests with a balance of 1 Health food, 1 Stamina food, and 1 Eitr food.

SETTLEMENT DOMAIN SYNERGY

SETTLEMENT BUFF KEEPER:
Supplies 3-course banquet rations to keep all workers operating at peak stamina.

FARMER VEGETABLE SINK:
Transforms raw carrots, turnips, onions, and barley into culinary dishes.

RANCHER MEAT CONSUMER:
Cuts raw meats into sausages, pies, and platters with zero waste.

SCHEMA 5: CULINARY TIMERS & MEAD FERMENTATION

Spit Timers • Cauldron Simmering • 2400s Mead Barrels • 3-Slot Banquets

STEP 01: SPIT ROAST TIMING



Roasts meats on open cooking spits; pulls at 20s to eliminate charcoal waste.

0-CHAR SPIT TIMER

STEP 02: CAULDRON SIMMERING



Combines vegetables, honey, and meats into high-tier HP and stamina stews.

CAULDRON RECIPES

STEP 03: STONE OVEN BAKING



Bakes bread dough, lox meat pies, and fish wraps in wood-fired bread ovens.

OVEN BAKERY

STEP 04: MEAD BASE BREWING



Simmers honey, berries, and bloodbags into potent medicinal base mixtures.

CAULDRON FERMENT BASE

STEP 05: 2400S MEAD AGING

Ferments mead bases in oak casks for 2400 seconds (2 in-game days).

CYCLE: 2400 SECONDS

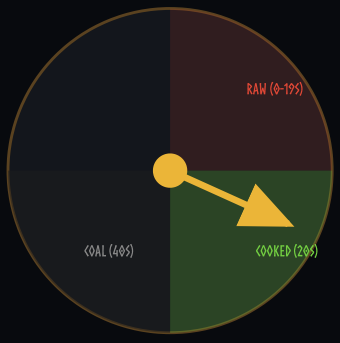
STEP 06: LARDER RATIONING



Sorts rations into cold storage chests tagged with [Food] and [Mead].

TAG: [FOOD] / [MEAD]

KITCHEN HEARTH & MEAD FERMENTATION CYCLE SPECIFICATION



SPIT FLIP AI (PULL AT 20.0S)



FERMENTER CELLAR

2400S (2 VALHEIM DAYS) • 6 POTIONS

CULINARY LAWS & TIMINGS

- F23 SPIT ROASTING: Pulls meats instantly at 20s (0% burn).
- F24 CAULDRON STEWS: Simmers deer, sausages, and turnip stews.
- F25 STONE OVEN: Bakes bread loaves and lox meat pies.
- F26 FERMENTER MEAD: Ages frost, poison, and healing meads.
- F27 LARDER SORTING: Routes cooked food to [Food] boxes.
- 3-SLOT DIET LAW: 1 HP + 1 Stamina + 1 Eitr maximizes stats.
- ZERO FOOD SPOILAGE: Cooked food never rots in chest storage.
- CAULDRON EXTENSION: Level 4 Cauldron unlocks Serpent Stew.
- AUTO-LOGGING: Feeds wood to oven fire autonomously.

PROVISIONS & HEARTH SYNERGIES

- RANCHER MEAT INGESTION: Consumes raw boar, wolf, and lox.
- FARMER PRODUCE PIPELINE: Converts raw crops into dishes.
- FISHER FILLET INTEGRATION: Uses raw fish for Fish Wraps.
- SENTRY MEAD RATIONING: Supplies healing mead to watchtowers.
- CHIMNEY VENTILATION: Maintain roof vents to prevent smoke.
- HAULER CONDUIT: Empties oven and cauldons to central vaults.
- COLD STORAGE: Keep larders roofed for 100% rest buffs.
- FERMENTER ROTATION: Run 3 fermenters to stagger mead yields.
- NOCTURNAL SHUTDOWN: Extinguishes spit flames during night.

CHEF'S TIMER LAW: Worker AI pulls roasting meat at exactly 20.0 seconds, yielding perfectly cooked rations with 0% charcoal loss.

PART VII: THE CRAFTSMAN (MASTER BUILDER & MASON)

CONSTRUCTION • REPAIR

LOADOUT: 

YIELDS & STATIONS: 

AUTONOMOUS AI WORK-CYCLE

1. PERIMETER DAMAGE SURVEY:
Patrols walls, palisades, and gates, identifying structures below 100% health.

2. ZERO-COST TIMBER REPAIRS:
Swaps to Hammer to repair wooden structures with zero material cost.

3. MASONRY FORTIFICATION:
Maintains stone bastions, trenches, and towers using the Stonecutter bench.

4. TOOL & GEAR RESTORATION:
Repairs weapons and armor in camp chests whenever durability drops below 10%.

ACTIVE DOMAIN CAPABILITIES

[F28] PERIMETER WALL MAINTENANCE:
Inspects and repairs damaged barricades, doors, and palisades.

[F29] ZERO-COST TIMBER RESTORATION:
Applies hammer repairs to wooden building components at no cost.

[F30] STONE & MASONRY REPAIRS:
Restores stone bastions, arches, and high towers using the Stonecutter.

[F31] WORKSTATION RESTORATION:
Restores durability to all player and worker tools stored in repair boxes.

HUGIN'S ADVICE & QUIRKS

STATION COVERAGE:
Ensure Workbenches and Stonecutters cover your entire perimeter wall to allow 100% repairs.

WEATHER DECAY DEFENSE:
Craftsmen continually restore rain damage on un-roofed wooden beams and stairs.

CHEST REPAIR LOOP:
Leave damaged tools in chests tagged [Repair]; the Craftsman restores them automatically.

SETTLEMENT DOMAIN SYNERGY

UNIVERSAL TOOL INTEGRITY:
Restores picks, axes, bows, and hoes for all other 9 worker roles.

SENTRY TOWER FORTIFICATION:
Maintains guard platforms, parapets, and palisades during enemy raids.

LUMBERJACK & MINER ALLIANCE:
Uses local timber and stone reserves to maintain impenetrable defense works.

TI: PART VIII: THE FISHER (COASTAL ANGLER & HARVESTER)

MARITIME • SEAFOOD

LOADOUT: 

YIELDS & STATIONS: 

AUTONOMOUS AI WORK-CYCLE

1. SHORELINE RECONNAISSANCE:
Walks coastline edges to identify fish schools in tidal waters.

2. CASTING & TENSION RETRIEVAL:
Casts baited lines; manages reel tension to land raw fish.

3. PIER FILLET STATION:
Cleans and fillets catches on dock tables, removing fish heads and entrails.

4. COLD STORAGE DEPOSITION:
Stores clean raw fish in barrels tagged [Fish] for the kitchen.

ACTIVE DOMAIN CAPABILITIES

[F32] COASTAL & DEEP ANGLING:
Casts into coastal waters to hook Perch, Pike, Trout, and Tuna.

[F33] REEL TENSION MANAGEMENT:
Successfully lands fish without snapping lines or running out of stamina.

[F34] DOCK FILLETING & STORAGE:
Cleans catches into raw fish units and routes them to cold storage larders.

HUGIN'S ADVICE & QUIRKS

DOCK CONSTRUCTION:
Build a wooden fishing pier extending 6m into deep water for higher-tier fish species.

BAIT SOURCING:
Keep a chest tagged [Bait] supplied with regular and mossy fishing bait on the pier.

RAIN FISHING BONUS:
Fish spawn rates increase during stormy weather; fishers double their yields.

SETTLEMENT DOMAIN SYNERGY

CHEF'S FISH WRAP SUPPLY:
Supplies fish fillets to the Cook for endgame Fish Wraps (+60 stamina).

MERCHANT COIN CYCLE:
Fish catches can be traded with Haldor or crafted into trophy mounts.

COASTAL SECURITY VIGIL:
Observes maritime approaches, warning Sentries of swimming threats.

PART IX: THE HAULER (LOGISTICS PORTER & QUARTERMASTER)

TRANSPORT • FREIGHT

LOADOUT: 

YIELDS & STATIONS: 

AUTONOMOUS AI WORK-CYCLE

1. FIELD VACUUM SWEEP:
Scans base grounds; vacuums dropped timber chunks, loose rocks, and stray monster loot.

2. CART HITCHING & LOADING:
Hitches to nearby wooden carts, filling mobile beds with heavy ores and stones.

3. SIGN BRACKET LOGISTICS:
Paths through depots, sorting gathered stacks into chests marked with [Item] tags.

4. SUPPLY DEPOT BALANCING:
Equalizes inventory stacks across multiple overflow boxes to prevent bin clogs.

ACTIVE DOMAIN CAPABILITIES

[F33] SETTLEMENT GROUND VACUUM:
Sweeps all loose dropped items within 25m, preventing visual clutter and lag.

[F34] TWO-WHEELED CART HAULING:
Hitches to and maneuvers heavy carts to transport up to 1000kg of raw materials.

[F35] BRACKET CONTAINER ROUTING:
Interprets sign tags to sort mixed items into designated containers with 100% accuracy.

[F36] INTER-DOMAIN DEPOT CONDUIT:
Relocates processed goods from production sheds to central stronghold vaults.

HUGIN'S ADVICE & QUIRKS

ROADWAY PLANING:
Use a Hoe to flatten 2m wide transit roads; haulers move 40% faster on stone paths.

STAIRWAY RAMPS:
Haulers cannot pull carts up standard steep stairs; build 26-degree roof or wood ramps.

SIGN SYNTAX PRECISION:
Brackets must match item names: [Wood], [Stone], [Ore], [Coal], [Produce], [Meat].

SETTLEMENT DOMAIN SYNERGY

THE VASCULAR BASE SYSTEM:
Connects all 9 production domains, freeing civilian workers from walking travel.

MINER CART INTEGRATION:
Empties mining pit chests directly, keeping excavators continuously swinging pickaxes.

FOUNDRY FEEDER LINK:
Transfers fuel and ores from distant yards directly into smelter hopper bins.

SCHEMA 6: GROUND VACUUM & FREIGHT

SCHEMA 6: GROUND VACUUM & FREIGHT ROUTING [TAGS]

25m Field Vacuum • 1000kg Cart Freight • Sign Tag Regex • Multi-Box Balancing

STEP 01: 25M GROUND VACUUM



Sweeps settlement ground; vacuums loose wood, stone, resin & drops.

SPHERE: 25M

STEP 02: CART HITCHING & LOAD



Hitches to 2-wheeled wooden carts; loads heavy ores, metals & stones.

CAPACITY: 1000 KG

STEP 03: WEIGHT AUDIT PASS



Monitors cart cargo weight to maintain 100% hauling mobility on ramps.

MASS BALANCE

STEP 04: CHEST BALANCING



Equalizes item stacks across multiple overflow boxes to prevent clogs.

LOAD BALANCING

STEP 05: SIGN TAG REGEX



Reads sign tags above chests (e.g. [Wood], [Ore]); routes cargo with 100% accuracy.

REGEX: [ITEM]

STEP 06: STRONGHOLD VAULT ROUTE



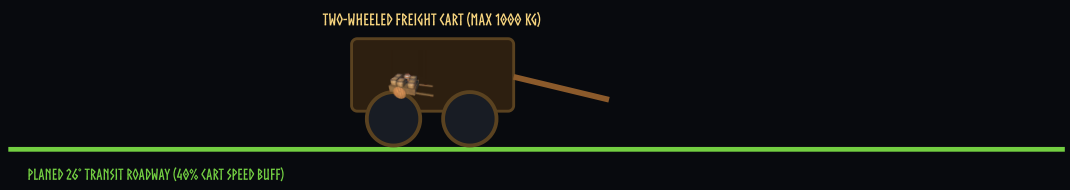
Delivers refined ingots, armors & food to fortified interior vaults.

SECURE CITADEL

FREIGHT DEPOT ARCHITECTURE & CONTAINER [TAG] REGEX

SIGN TAG REGEX ENGINE: [([A-ZA-Z0-9_-]+)]

Workers parse bracketed sign text above containers to route mixed inventory with 100% precision.



LOGISTICS ROUTING RULES

- F33 GROUND VACUUM: Clears all dropped items within 25m.
- F34 CART HAULING: Hitches to wagons; relocates 1000kg.
- F35 TAG SORTING: Case-insensitive regex bracket matching.
- F36 DEPOT CONDUIT: Relocates goods to central vaults.
- ROADWAY SPEED BUFF: Carts move 40% faster on stone paths.
- 2M DOORWAY CLEARANCE: Carts require 2m wide openings.
- ZERO ITEM LOSS: Dropped materials never despawn in camps.
- MULTI-CART SUPPORT: Haulers switch carts when full.
- COLLISION EVASION: Paths around obstacles cleanly.

CONTAINER TAG SYNTAX TABLE

- [WOOD] & [FINEWOOD]: Round logs, core wood & fine wood.
- [ORE] & [STONE]: Copper, tin, iron, silver & rocks.
- [PRODUCE] & [SEED]: Vegetables, barley, flax & seeds.
- [MEAT] & [FEED]: Raw cuts, neck tails & animal feed.
- [INGOT] & [TOOLS]: Refined metal bars & forged gear.
- [REPAIR]: Damaged items routed for zero-cost repairs.
- [ARROWS]: Ammunition bins for watchtower sentries.
- OVERFLOW FALLBACK: Builds 50x piles if boxes are full.
- CASE INSENSITIVE: [wood] matches [WOOD] perfectly.

Sign **SIGN ROUTING LAW: Surround item names with square brackets e.g. [Wood], [Ore], [Ingot]. Hauler AI automatically routes cargo to matching containers.**

PART X: THE SENTRY (STRONGHOLD GUARD & PATROL)

SECURITY • DEFENSE

LOADOUT: 

YIELDS & STATIONS: 

AUTONOMOUS AI WORK-CYCLE

1. WATCHTOWER SENTRY VIGIL:
Mans high defensive battlements, keeping 360-degree vision over outer wilderness.

2. THREAT RECOGNITION:
Identifies Greydwarf scouts, Trolls, and nocturnal predators up to 40m away.

3. RANGED HARASSMENT ARCS:
Draws bows to suppress enemy targets with fire, flint, and needle arrow volleys.

4. PERIMETER MELEE INTERCEPT:
Swaps to shield and blade to engage breaches before monsters can touch civilian workers.

ACTIVE DOMAIN CAPABILITIES

[F37] PERIMETER PATROL & VIGIL:
Walks ramparts and gatehouses on programmed defensive patrol paths.

[F38] TACTICAL RANGED SUPPRESSION:
Fires aimed bow volleys to neutralize incoming raid forces before they reach gates.

[F39] AGGRO INTERCEPTION & SHIELD WALL:
Draws hostile focus away from civilian workers using aggressive melee blocking.

[F40] POST-RAID DROP SWEEPING:
Secures combat spoils (eyes, resin, bones, trophies) and dumps them in barracks.

HUGIN'S ADVICE & QUIRKS

ELEVATED POSTS:
Station Sentries in covered watchtowers so they gain height advantage and stay sheltered from rain.

ARROW AMMUNITION BINS:
Place a [Arrows] chest in the guard tower to provide infinite autonomous quiver refills.

LIGHTING ADVANTAGE:
Place standing torches along perimeter fences to expose nocturnal attackers to sentry fire.

SETTLEMENT DOMAIN SYNERGY

CIVILIAN WORKER DEFENSE:
Guarantees Farmers, Lumberjacks, and Miners can work without fear of random ambushes.

CRAFTSMAN ARMORY TIE-IN:
Receives freshly forged armor, bows, and shields directly from the Stronghold armorer.

RAID SPOILS SUPPLY:
Provides constant Grey Dwarf eyes and trophies to Cook and Craftsman for advanced recipes.

SCHEMA 7: CITADEL DEFENSE & SIGHTLINES

40m Threat Acquisition • Intersecting Sightlines • Shield Wall Intercept • Spoils Pass

STEP 01: WATCHTOWER VIGIL



Mans elevated battlements; scans wilderness with 360-degree perimeter vision.

VIGIL: 360°

STEP 02: 40M THREAT ACQUISITION



Identifies Greydwarfs, Trolls & nocturnal predators up to 40m away.

TARGET RANGE: 40M

STEP 03: AIMED RANGED VOLLEYS



Draws bows to suppress enemy targets with fire, flint, and needle arrows.

ARROW SUPPRESSION

STEP 04: MELEE INTERCEPT



Swaps to blade & shield to engage breaches before monsters touch workers.

MELEE ENGAGEMENT

STEP 05: SHIELD WALL ENGAGE



Engages breaches with Banded Shield; draws hostile aggro away from civilians.

AGGRO INTERCEPTION

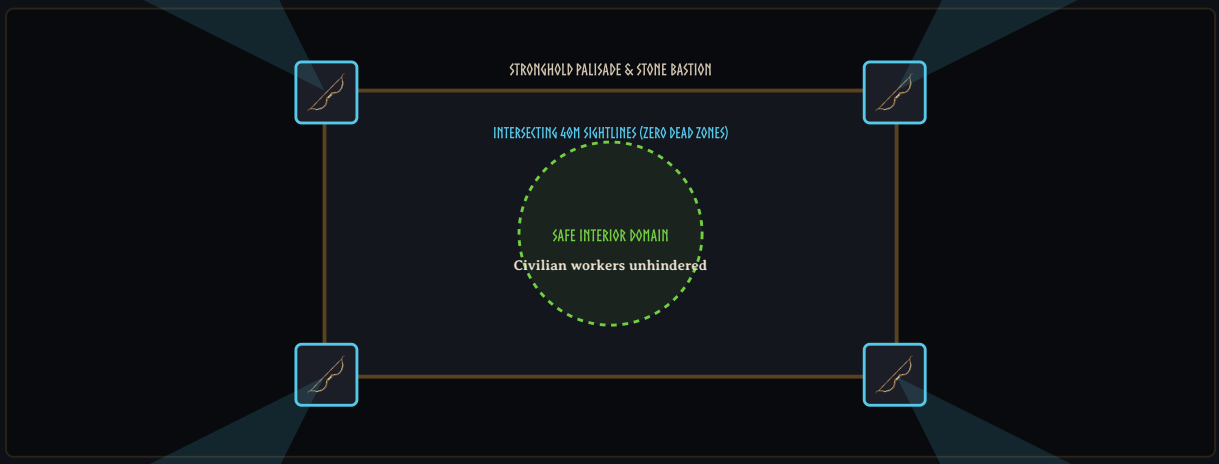
STEP 06: SPOILS SWEEP PASS



Secures drops (eyes, resin, bones, trophies); routes loot to barracks.

POST-RAID SWEEP

CITADEL DEFENSE GEOMETRY & INTERSECTING SIGHTLINES



PERIMETER DEFENSE SPECIFICATIONS

- F37 PERIMETER PATROL: Walks ramparts on programmed paths.
- F38 RANGED SUPPRESSION: Aimed bow volleys neutralize threats.
- F39 AGGRO INTERCEPT: Banded Shield blocks protect civilians.
- F40 POST-RAID SWEEPER: Secures combat trophies & monster drops.
- 40M TARGET RANGE: Acquires nocturnal predators before breaches.
- HEALTH SAFETY RETREAT: Disengages when HP < 30% to heal.
- NIGHT VIGIL: Sentries remain awake through night shifts.
- ARMORY REFILL: Auto-pulls arrows from [Arrows] chests.
- CITADEL WALLS: Craftsman keeps palisades at 100% health.

LOGISTICS & ARMORY CONDUITS

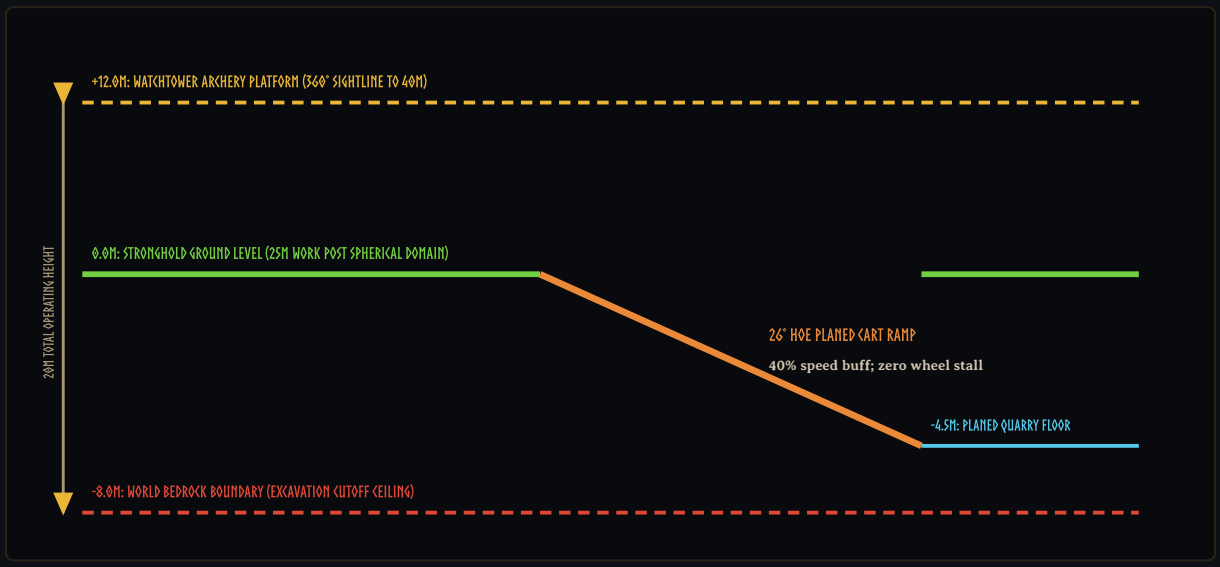
- ARROW AMMUNITION BOXES: Auto-refills from [Arrows] chests.
- CRAFTSMAN ARMORY FORGE: Receives repaired shields & swords.
- WALL INTEGRITY ALLIANCE: Craftsman restores gate damage.
- RAID SPOILS SUPPLY: Delivers eyes and resin to kitchen & forge.
- PERIMETER BRAZIERS: Standing torches eliminate stealth bonuses.
- COVERED TOWER ROOFS: Protects sentries from rain & drakes.
- CHOKEPOINTS: Channel attackers through 2m fortified gates.
- LEASH RADIUS: Sentries never pursue beyond 40m perimeter.
- STAGGER MITIGATION: Banded shield absorbs troll club blows.

DEFENSE GEOMETRY LAW: Position watchtowers at diagonal stronghold corners so archer sightlines cross-cover all gate approaches without dead zones.

SETTLEMENT SPATIAL ARCHITECTURE & 3D ELEVATION BLUEPRINT

Vertical Elevation Bounds • 26° Ramp Physics • Navmesh Grids • Comfort Radii

THREE-TIER VERTICAL ELEVATION PROFILES & CART TRACTION



NAVMESH SNAPPING & SETTLEMENT SPATIAL PARAMETERS

SPATIAL SNAPPING & NAVMESH LAWS

- 25M WORK SPHERE: Work posts radiate a 25m sphere.
- 4M POST SNAPPING: Work posts snap to 4m wood grid.
- 2M DOORWAYS: Doors must be 2m wide for cart clearance.
- STAIRWAY RAMPS: Replace steep 45° stairs with 26° ramps.
- 8M CHEST INGESTION: Smelter chests must sit <= 8m away.
- ZERO RELOCATION PENALTY: Break & move posts anytime.
- NAVMESH REBUILD: Press F5 and enter worker_reset.
- ROOF COVERAGE: 100% shelter stops rain durability rot.
- WORKBENCH SPHERE: 1 Workbench covers 20m building zone.
- CHIMNEY CLEARANCE: 2x2m smoke vents prevent choke.
- CHOPPING BLOCK: Station upgrade required for Bronze tools.
- ANVIL EXPANSION: Level 2 Forge enables Iron tool repair.
- MAXIMUM WORKERS: Caps at 12 active workers per 64m zone.

COMFORT, SHELTER & REST CYCLES

- SHELTER BUFF: Roof + walls provide 100% rest recovery.
- DRAGON BED: Assign workers to beds for instant stamina.
- HEARTH PROXIMITY: 14m warmth radius stops freezing.
- NOCTURNAL SLEEP: Civilian workers sleep 21:00 to 06:00.
- SENTRY VIGIL: Sentries remain awake through night shifts.
- HYPOTHERMIA IMMUNITY: 3-slot food sets negate cold status.
- CAMP INTEGRITY: 1 upgraded Workbench covers 20m building.
- COMFORT LEVEL 17: Dragon beds & rugs yield 24min buff.
- BRAZIER LIGHT: Standing iron braziers warm 10m radius.
- STORM BEHAVIOR: Civilians retreat indoors during storms.
- REPAIR DISENGAGE: 10% durability threshold initiates retreat.
- FOOD AUTO-FEED: Workers consume food from larder chests.
- HUGIN NOTIFICATIONS: F5 status prints worker diagnostics.

BLUEPRINT LAW: Maintain strict 26° slopes on all roadways and mining pit ramps. Steeper grades break cart physics and cause workers to drop hitches.

ADMIN CONTROLS: 

DIAGNOSTIC STATUS: 

TACTICAL AI TROUBLESHOOTING

WORKER STUCK IN PLACE (PATHING DEADLOCK)

Use a Hoe to smooth abrupt cliff edges. Ensure doorways are at least 2m wide and stairs have no steep obstructions.

WORKER IDLE ("NO WORK AVAILABLE")

Confirm target nodes (rocks, crops, trees) are within the 25m sphere and required tool tiers are available in chests.

CONTAINER FULL DEADLOCK

Workers halt extraction if matching chests are full. Enable 50x resource piles or add extra chests tagged with [Tag].

WORKBENCH LEVEL INSUFFICIENT

High-tier tool repairs require upgraded station levels. Place Anvils, Coolers, and Chopping Blocks nearby.

CONFIGURATION MATRIX (VALHEIMWORKERS.CFG)

SCANINTERVALSECONDS = 2.5

Evaluation tick frequency in seconds. Lower values increase responsiveness; higher values optimize potato CPUs.

MAXWORKERSPERZONE = 12

Caps total concurrent autonomous workers permitted inside a single active 64m world zone to protect server FPS.

PATHINGMESHDETAIL = HIGH

Controls navmesh resolution around player structures, doors, and complex terrain modifications.

AUTODEPOSITOVERFLOW = TRUE

Allows Lumberjacks and Miners to erect physical 50x wood and stone piles when storage boxes are capped.

TOOLDURABILITYTHRESHOLD = 0.10

Percentage (10%) at which workers disengage from tasks to seek repair at a Workbench or Forge.

HUGIN'S GOLDEN RULES OF BASE DESIGN

CENTRAL LOGISTICS HUB

Place Hauler depots in the geographic center of your base to minimize transfer distance between quadrants.

RAMPED ROADWAYS

Pave dirt roads into flat stone or leveled soil paths; carts move 40% faster and rarely lose traction on ramps.

OVERLAPPING WATCHTOWERS

Build guard towers at perimeter corners so Sentry bow sightlines intersect, eliminating all dead zones.

CONSOLE COMMANDS (F5)

WORKER_DEBUG ON/OFF

Toggles visual debug rendering of the 25m operating sphere, task raycasts, and pathing nodes.

WORKER_RESET

Forces all nearby workers to drop current tasks, re-scan inventory, and re-bind to valid work posts.

WORKER_STATUS

Prints a full diagnostic report of nearby worker inventories, stamina, and current AI states to the console.